



Client-Conditional Federated Learning

via Local Training Data Statistics

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The problem: heterogeneity breaks FL's promise

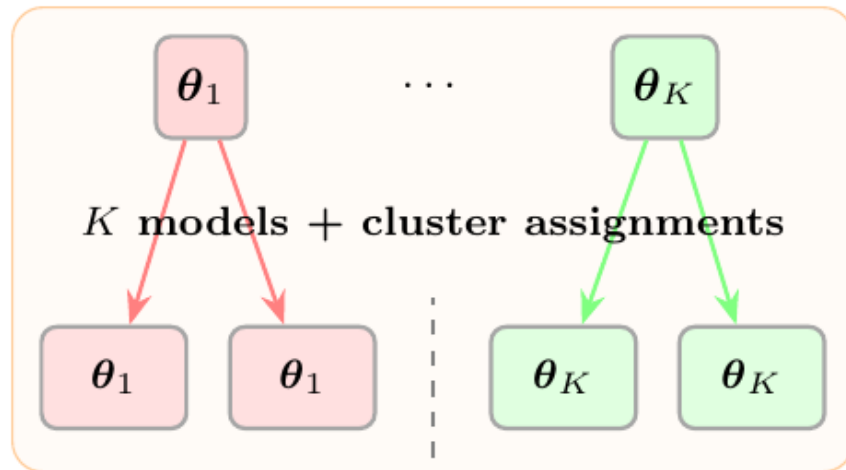
- FL works when collaboration beats going it alone — data heterogeneity breaks that.
- Four flavours: label (E1), covariate (E2), concept (E3), combined (E4) shift.
- FedAvg compromises across all local objectives — and collapses.

CIFAR-10, K: 2 → 10 clusters

73.5% → 17.2%

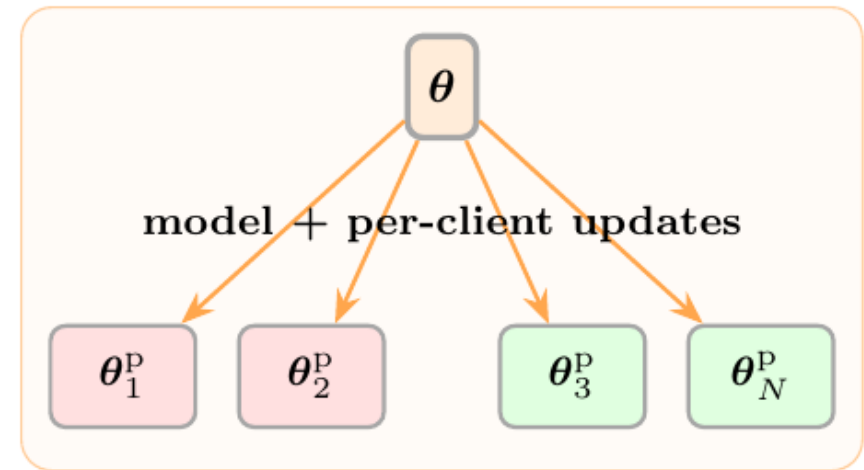
FedAvg accuracy under label shift

Existing approaches discover client relationships



K cluster models $f(\mathbf{x}; \theta_k)$
— iterative discovery

Clustered — IFCA, CFL



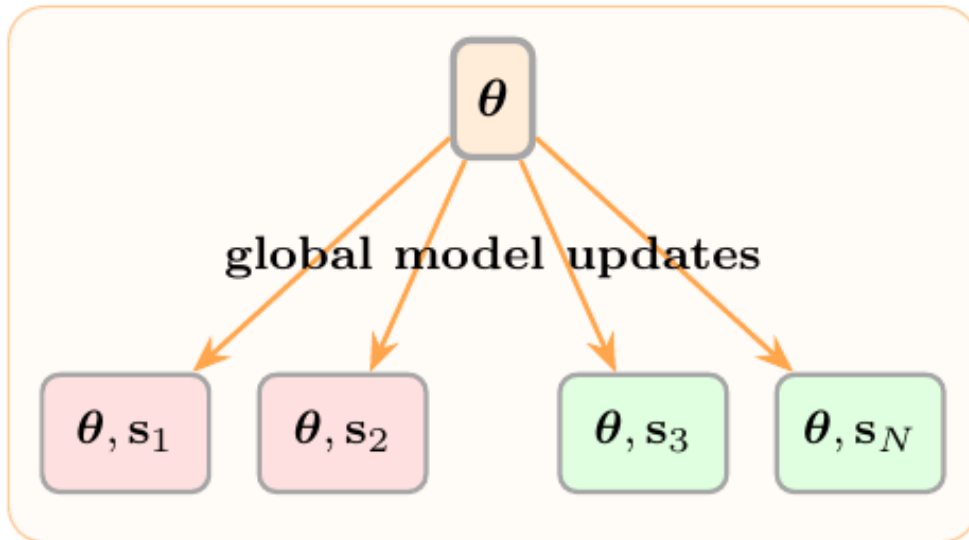
N personal models $f(\mathbf{x}; \theta_i^p)$
— per-client fine-tuning

Personalized — Ditto, Per-FedAvg

Decentralized methods (DAC) discover the collaboration graph instead.

Shared weakness: all infer similarity from model behaviour — unreliable under sparsity, can disclose information, and assume a few discrete clusters.

Our idea: characterize, don't discover



one shared model $f(\mathbf{x}; \theta, s_i)$
— zero extra communication

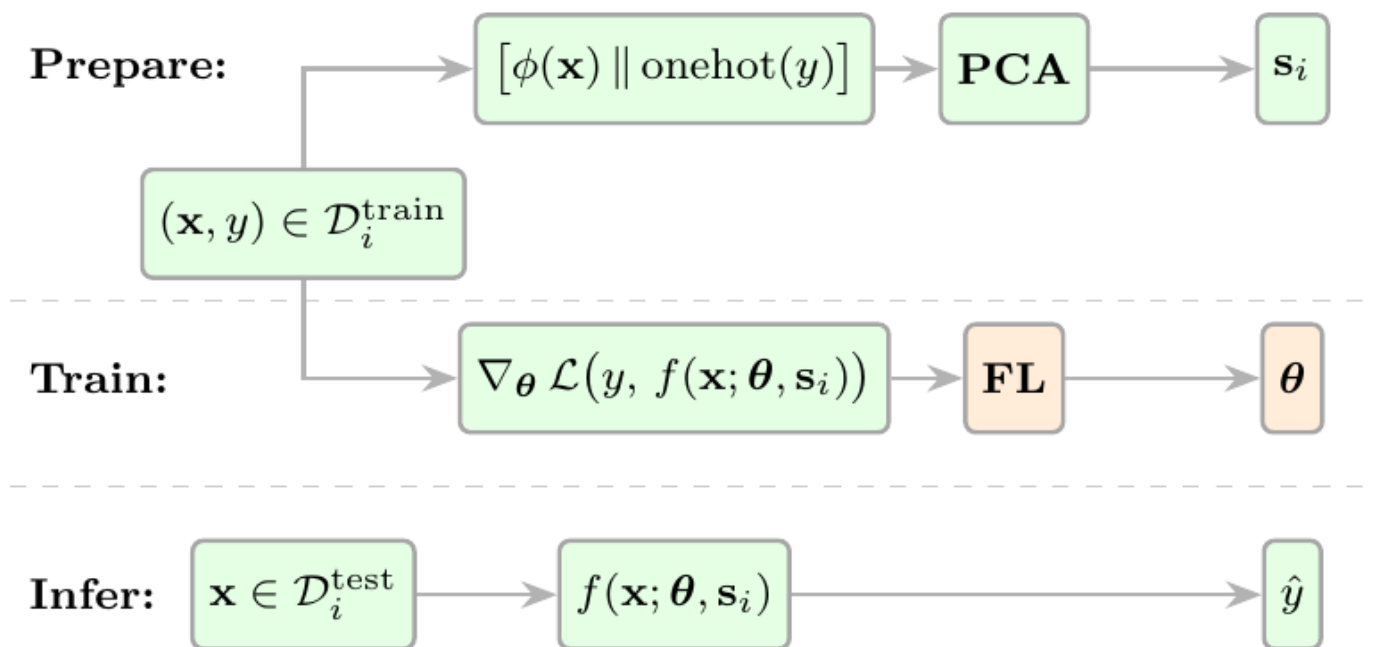
Conditional (Ours)

- Each client computes a compact PCA-eigenvalue fingerprint of its own data.
- One global model is conditioned on it.
- No separate models, no cluster discovery.
- **No extra communication or disclosure beyond standard FL.**

Method: fingerprint + conditioning

1. Fingerprint $Z = [\phi(\mathbf{x}) \parallel \text{onehot}(y)] \rightarrow \text{top-32 PCA eigenvalues} \rightarrow \mathbf{s}$. Magnitude of variation; rotation-invariant; sensitive to all three shifts; computed locally.

2. Conditioning Concatenate $\mathbf{s}$ with flattened CNN features before the FC layers (Conv $\rightarrow$ Flatten $\rightarrow [\mathbf{f} \parallel \mathbf{s}] \rightarrow \text{FC} \rightarrow \text{class}$). Conv shared; FC adapts. Adds $< 1\%$ parameters; same communication pattern as FedAvg.



Experimental setup

- **97 configurations stress-testing every heterogeneity type.**
- 4 types (E1–E4) × 4 datasets (MNIST, FMNIST, CIFAR-10/100).
- 7 baselines: Local, FedAvg, Gossip, IFCA, DAC, Ditto, Oracle.
- Sparsity sweep: 6,000 → 200 samples / client.

The bar to beat

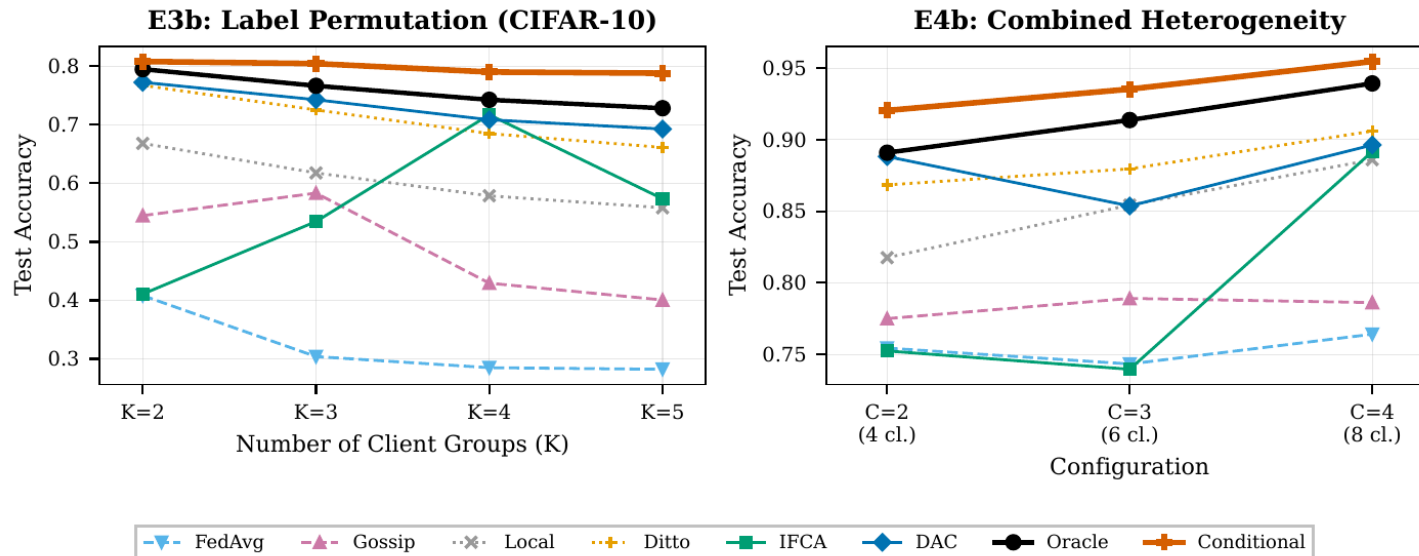
Oracle trains one model per true cluster — it knows the answer. The ceiling for clustering methods.

Claim 1 — matches or beats Oracle across all types

Exp.	Representative config	FedAvg	Oracle	Cond	Best other
E1	CIFAR-10 superclass K=5	.325	.927	.929	.906 (IFCA)
E2a	CIFAR-10 subclass K=3	.943	.979	.981	.972 (DAC)
E2b	FMNIST rotate K=4	.789	.900	.895	.868 (DAC)
E3a	FMNIST semantic, rich	.671	.950	.950	.943 (IFCA)
E3b	CIFAR-10 permute K=2	.408	.795	.808	.773 (DAC)
E4a	MNIST+FMNIST, rich	.939	.957	.955	.953 (IFCA)
E4b	CIFAR-10 combine C=3	.743	.914	.935	.880 (Ditto)

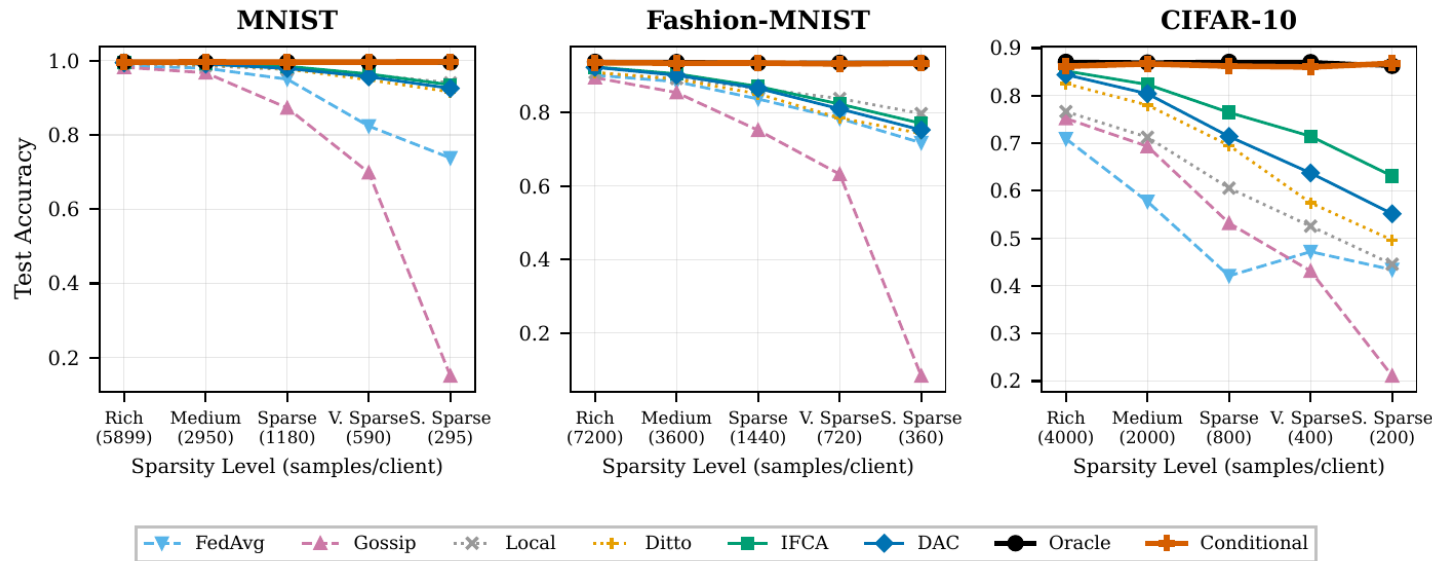
- With no cluster knowledge, Conditional matches Oracle within ~1% — and tops every non-Oracle baseline.
- **Across 97 configs it matches or beats Oracle in 95 (98%).**

Claim 2 — beats Oracle on combined heterogeneity



- A continuous 32-d fingerprint encodes multiple axes; a single cluster ID cannot.
- **E4b combined: +1.5 to +2.9%**
- **E3b permute (CIFAR-10): up to +6%**

Claim 3 — uniquely sparsity-robust



- 20× less data per client (CIFAR-10: 4,000 → 200):
- **Conditional & Oracle: flat (+0.7%, -0.9%).**
- IFCA -26%, DAC -35%, Ditto -40%, Gossip -72%.
- Fingerprint is precomputed; model trains on pooled data.

Takeaways

- **Characterize > discover.**
- Condition one shared model on local PCA statistics.
- No extra communication rounds or disclosure; only a $< 1\%$ larger model; no cluster estimation.
- Matches Oracle across all heterogeneity types; beats it on multi-dimensional heterogeneity; uniquely sparsity-robust.
- Best when heterogeneity type is unknown, data is scarce, comms are limited.

Backup — E1 label shift, full K sweep

	MNIST				Fashion-MNIST				CIFAR-10			
Method	K2	K3	K5	K10	K2	K3	K5	K10	K2	K3	K5	K10
FedAvg	.986	.978	.933	.522	.902	.886	.745	.412	.735	.554	.325	.172
IFCA	.995	.995	.999	1.00	.922	.932	.945	1.00	.847	.766	.906	1.00
DAC	.995	.995	.998	1.00	.922	.931	.950	1.00	.844	.859	.898	1.00
Oracle	.996	.996	.999	1.00	.937	.945	.962	1.00	.861	.886	.927	1.00
Cond	.996	.996	1.00	1.00	.936	.942	.964	1.00	.860	.875	.929	1.00

FedAvg collapses as K grows; Conditional tracks Oracle within 1.1% everywhere.